

Exact Schur–Extension and 7-Adic Principal-Coordinate Structure in a Finite Integer System

Computational Audit Report

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Abstract

This paper records an exact computational audit of a structured integer matrix together with a prime-local analysis of its Schur extension. The mod-2 computation isolates a rank-one Schur correction and a one-dimensional row/column quotient. The distinguished adjugate line lies in the left/right kernels of the 12×12 core. Independently, a pair of terminal source values determines a projective 7-adic ratio. After writing the exponent as $k = k_1 + 6m$, the orbit equation becomes a principal-coordinate equation in the unit $729 = 3^6$. The paper proves the local linearization behind the observed affine Hensel law and gives an exact logarithmic coordinate formula $m = \log(\rho/(2 \cdot 3^{k_1}))/\log(729)$ in the appropriate 7-adic domain. Exact computations through modulus 7^{16} and a complete audit over all nonzero residues of $\mathbf{F}_7$ support the stated mechanism. The factor 17 is shown, by counterfactual source scaling, to affect Schur amplitude without changing the projective 7-adic exponent coordinate.

1 Introduction

The experiments summarized here arose from an exact audit of a 14×14 integer matrix M with a distinguished 12×12 retained block F_{ret} , a 12×12 core block A , and a 2×2 boundary block

$$H = \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}, \quad \det H = 2.$$

The computational question was initially whether the rank drop between the retained operator and the Schur-corrected core was an artifact of a particular representation. A sequence of exact mod-2 audits instead revealed a stable one-dimensional quotient together with a rank-one Schur correction. A separate 17-adic audit localized the common factor 17 to two terminal source values and showed that common rescaling preserves the projective source ratio.

The later 7-adic experiments uncovered a second structure. Two primitive terminal values

$$q_1 = 29,144,191, \quad q_3 = 24,794,967$$

determine the unit ratio

$$\rho = \frac{q_1}{q_3}.$$

The exponent equation is

$$2 \cdot 3^k \equiv \rho \pmod{7^E}.$$

For the observed data, $\rho \equiv 6 \pmod{7}$, hence $k_1 \equiv 1 \pmod{6}$. Writing $k = 1 + 6m$ gives

$$2 \cdot 3^k = 6 \cdot 729^m.$$

This change of coordinate explains the persistent slope-one Hensel law and leads to a closed logarithmic description of the principal coordinate.

The purpose of the present paper is therefore not to reproduce every experiment, but to organize the exact facts into a theorem-oriented account. All matrix and modular rank statements cited below were evaluated over the exact finite field $\mathbf{F}_2$ or by exact integer arithmetic; the 7-adic computations use modular exponentiation and truncated p -adic series with rational arithmetic.

2 Exact matrix and Schur structure

Partition the matrix in the form

$$M = \begin{pmatrix} F_{\text{ret}} & B \\ C & H \end{pmatrix},$$

where $F_{\text{ret}}, A \in M_{12}(\mathbf{Z})$, $B \in M_{12 \times 2}(\mathbf{Z})$, $C \in M_{2 \times 12}(\mathbf{Z})$, and $H \in M_2(\mathbf{Z})$. Over $\mathbf{Q}$, H is invertible and the Schur complement is

$$A = F_{\text{ret}} - BH^{-1}C.$$

The exact computational identity used in the audits was also written as

$$K = A - F_{\text{ret}} = -BH^{-1}C.$$

Proposition 2.1 (Exact mod-2 ranks). *For the audited matrix,*

$$\text{rank}_{\mathbf{F}_2}(M) = 4, \quad \text{rank}_{\mathbf{F}_2}(F_{\text{ret}}) = 4, \quad \text{rank}_{\mathbf{F}_2}(A) = 3, \quad \text{rank}_{\mathbf{F}_2}(K) = 1.$$

Hence

$$\text{rank}(F_{\text{ret}}) - \text{rank}(A) = 1.$$

Proposition 2.2 (Row and column quotient geometry). *Over $\mathbf{F}_2$,*

$$\dim(\text{Row } F_{\text{ret}} \cap \text{Row } A) = 3, \quad \dim(\text{Row } F_{\text{ret}} / \text{Row } A) = 1,$$

and analogously,

$$\dim(\text{Col } F_{\text{ret}} \cap \text{Col } A) = 3, \quad \dim(\text{Col } F_{\text{ret}} / \text{Col } A) = 1.$$

Moreover K contributes exactly those quotient directions:

$$\text{Row } F_{\text{ret}} = \text{Row } A \oplus \text{Row } K,$$

$$\text{Col } F_{\text{ret}} = \text{Col } A \oplus \text{Col } K.$$

The distinguished Schur correction has support

$$\text{supp}(K_{\text{mod } 2}) = \{(11, 0), (11, 3), (11, 5)\},$$

so it is a single nonzero row in the mod-2 core. The corresponding row vector is

$$[1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0].$$

Proposition 2.3 (Adjugate kernel directions). *The normalized adjugate directions obtained from the Schur boundary have representatives*

$$a_{\text{row}} = [0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0],$$

$$a_{\text{col}} = [0, 0, 0, 1, 0, 1, 0, 1, 0, 1, 1, 1].$$

Over $\mathbf{F}_2$,

$$a_{\text{row}} \in \ker(A^T), \quad a_{\text{col}} \in \ker(A).$$

Since $\text{rank}(A) = 3$, both left and right nullities are 9.

3 The rank-one Schur correction and its arithmetic source

The boundary block has

$$H^{-1} = \frac{1}{2} \begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix},$$

so

$$K = -\frac{1}{2} B \operatorname{adj}(H) C.$$

Exact computation shows that $B \operatorname{adj}(H) \equiv 0 \pmod{2}$ and $\operatorname{adj}(H) C \equiv 0 \pmod{2}$, while $K \not\equiv 0 \pmod{2}$. The first nonzero 2-adic layer is obtained after dividing these intermediate factors by 2.

In the original boundary ordering, the first lifted left factor is a matrix unit modulo 2:

$$L_1 = \frac{B \operatorname{adj}(H)}{2}, \quad L_1 \equiv e_{11} e_1^T \pmod{2}.$$

Consequently,

$$K \equiv e_{11} C_0 \pmod{2},$$

where C_0 is the first terminal boundary row. The other boundary row contributes only at the next 2-adic layer.

The exact terminal source values are

$$q_1 = 495451247, \quad q_3 = 421514439$$

in the unnormalized system, with

$$\gcd(q_1, q_3) = 17.$$

After division by 17 they become the primitive pair used below:

$$q'_1 = 29144191, \quad q'_3 = 24794967.$$

The terminal Schur row is exactly

$$(q'_1 - 2q'_3, q'_1 - 6q'_3, q'_1 - 18q'_3)$$

inserted into the active basis coordinates.

4 Localization of the factor 17

The exact 17-adic audit found

$$\gcd(C) = 17, \quad \gcd(K) = 17,$$

while B and H have no 17-content. Dividing gives primitive matrices $C/17$ and $K/17$, and recomputation of the Schur correction from $C/17$ reproduces $K/17$ exactly.

At the source-table level, exactly two q -values are divisible by 17, namely the two terminal values. All other recorded q -values are 17-adic units. Thus the observed 17-content is source-localized rather than a global lattice factor.

The strongest counterfactual is exact common rescaling. For $s \in \{2, 3, 5, 17, 17^2\}$,

$$(sq_1, sq_3)$$

has exactly the same ratio as (q_1, q_3) . The experiment verified that the entire 7-adic exponent sequence, including all lift digits, remains unchanged. In contrast, the terminal Schur source row scales linearly:

$$S(sq_1, sq_3) = sS(q_1, q_3).$$

For $s = 17$ this reproduces precisely the observed factor in the terminal Schur row.

Proposition 4.1 (Projective/content decoupling). *Common source scaling leaves the projective 7-adic exponent data invariant, while scaling the terminal Schur source row by the same scalar. One-sided source scaling changes the projective ratio and, in the tested control, changes both the mod-7 base exponent and the lift-slope sequence.*

5 The 7-adic orbit and principal coordinate

Let

$$\rho = \frac{q_1}{q_3} \in \mathbf{Z}_7^\times.$$

The exponent equation is

$$2 \cdot 3^k \equiv \rho \pmod{7^E}.$$

Since 3 has order 6 modulo 7, every nonzero residue $\rho \in \mathbf{F}_7^\times$ determines a unique residue class $k_1 \pmod{6}$ satisfying

$$2 \cdot 3^{k_1} \equiv \rho \pmod{7}.$$

Write

$$k = k_1 + 6m.$$

Then

$$2 \cdot 3^k = (2 \cdot 3^{k_1})729^m.$$

Thus the 7-adic exponent problem becomes a principal-unit problem in the coordinate m .

For the original dataset,

$$\rho \equiv 6 \pmod{7}, \quad k_1 = 1,$$

so

$$\rho = 6 \cdot 729^m$$

in the 7-adic sense.

6 Exact local linearization

The crucial identity is

$$729 = 1 + 728 = 1 + 7 \cdot 104, \quad 104 \equiv 6 \pmod{7}.$$

Lemma 6.1 (Principal-unit linearization). *For every integer $e \geq 1$ and every $t \in \mathbf{Z}$, one has*

$$729^{t7^{e-1}} \equiv 1 + t(729 - 1)7^{e-1} \pmod{7^{e+1}}.$$

Proof. Write $u = 729 - 1$, so $v_7(u) = 1$. Then

$$729^{t7^{e-1}} = (1 + u)^{t7^{e-1}}.$$

The linear term is

$$t7^{e-1}u,$$

which has valuation e . Every quadratic and higher binomial term has valuation at least $e+1$, because it contains u^2 together with the corresponding binomial coefficient contribution. Equivalently, this is the standard first-order principal-unit expansion modulo 7^{e+1} . Hence the displayed congruence follows. $\square$ $\square$

Theorem 6.2 (General Hensel-slope formula). *Let $\rho \in \mathbf{F}_7^\times$, choose the unique $k_1 \pmod{6}$ with*

$$2 \cdot 3^{k_1} \equiv \rho \pmod{7},$$

and write $k = k_1 + 6m$. At every Hensel lifting level, the normalized residual as a function of the next digit $t \in \mathbf{F}_7$ is affine,

$$R_e(t) = A_e + B(\rho)t,$$

with

$$\boxed{B(\rho) \equiv \rho \frac{729 - 1}{7} \equiv 6\rho \pmod{7}.}$$

In particular $B(\rho) \neq 0$ for every $\rho \in \mathbf{F}_7^\times$, so every lift digit is unique.

Proof. At a given level, replace m by

$$m + t7^{e-1}.$$

The orbit is multiplied by $729^{t7^{e-1}}$. By the lemma,

$$729^{t7^{e-1}} \equiv 1 + t(729 - 1)7^{e-1} \pmod{7^{e+1}}.$$

After normalization by 7^e , the change is therefore

$$\rho t \frac{729 - 1}{7} \pmod{7},$$

because the current orbit residue agrees with ρ modulo 7. Hence

$$B(\rho) \equiv \rho \frac{729 - 1}{7} \equiv 6\rho \pmod{7}.$$

Since $\rho \neq 0$ and $6 \neq 0$ in $\mathbf{F}_7$, the slope is a unit, so there is a unique digit solving the next congruence. $\square$ $\square$

Corollary 6.3. *For the original source ratio $\rho \equiv 6 \pmod{7}$,*

$$B(6) = 36 \equiv 1 \pmod{7}.$$

Thus the experimentally observed law

$$R_e(t) = A_e + t$$

is forced by the local algebra.

7 Complete finite-field verification

The general slope formula was tested for every nonzero residue $\rho \in \mathbf{F}_7^\times$. The complete table is

ρ	1	2	3	4	5	6
k_1	4	0	5	2	3	1
$B(\rho)$	6	5	4	3	2	1

For each of the six ratios, the seven possible Hensel lifts were enumerated at eight successive levels. In every case the residual was exactly affine, the slope agreed with 6ρ , the successful lift digit was unique, and the lifted exponent reproduced the prescribed ratio modulo the next power of 7.

This is a finite exact verification of the universal local mechanism over $\mathbf{F}_7^\times$.

8 The logarithmic principal coordinate

For the original ratio, $\rho/6 \equiv 1 \pmod{7^2}$, while $729 \equiv 1 \pmod{7}$. Hence both 7-adic logarithms converge:

$$\log(\rho/6), \quad \log(729).$$

The valuation audit gives

$$v_7(\log(\rho/6)) = 2, \quad v_7(\log(729)) = 1.$$

Therefore the quotient

$$\mu := \frac{\log(\rho/6)}{\log(729)}$$

is a well-defined 7-adic integer with $v_7(\mu) = 1$.

Theorem 8.1 (Logarithmic principal coordinate). *For the original source ratio,*

$$m = \frac{\log(\rho/6)}{\log(729)}$$

in $\mathbf{Z}_7$. Modulo 7^{11} , the logarithmic coordinate agrees exactly with the independently reconstructed Hensel coordinate. More generally, at every tested precision 7^E ,

$$m \equiv \frac{\log(\rho/6)}{\log(729)} \pmod{7^{E-1}}.$$

The equality was checked independently through $E = 16$. The reconstructed principal coordinate at precision 7^{16} was

$$m_{16} = 1,298,954,352,220,$$

with

$$m_{16} \equiv \frac{\log(\rho/6)}{\log(729)} \pmod{7^{15}}.$$

The corresponding exponent is

$$k_{16} = 1 + 6m_{16} = 7,793,726,113,321.$$

Proof. The exact identity $\rho/6 = 729^m$ in the relevant 7-adic unit neighborhood implies, by the homomorphism property of the 7-adic logarithm,

$$\log(\rho/6) = m \log(729).$$

Since $\log(729)$ is a nonzero 7-adic element, division gives the formula. The finite-precision agreement was verified independently by exact rational truncations of the logarithmic series and by modular Hensel reconstruction. $\square$ $\square$

9 Canonical base-7 digit expansion

The Hensel recurrence has the form

$$m_{e+1} = m_e + t_e 7^{e-1}, \quad t_e \in \{0, \dots, 6\}.$$

Consequently the tested sequence satisfies

$$m_E = \sum_{j=1}^{E-1} t_j 7^{j-1}.$$

For the original data the verified digits through precision 7^{12} are

$$(t_1, \dots, t_{11}) = (0, 2, 5, 0, 5, 5, 4, 0, 2, 3, 6).$$

The corresponding exact principal coordinates are

$$\begin{aligned} &0, 0, 14, 259, 259, 12264, 96299, 566895, \\ &566895, 12096497, 133157318, 1828008812. \end{aligned}$$

Direct integer base-7 conversion reproduces precisely the same digit strings at every tested precision.

Thus the Hensel process does not merely generate an arbitrary sequence of integers: it constructs the canonical base-7 expansion of the principal logarithmic coordinate.

10 High-precision exact audit

The theorem-oriented computations were extended through modulus 7^{16} . The resulting exponent sequence was reconstructed from the source ratio alone. The final exponent was

$$7,793,726,113,321,$$

and every lift was independently checked against

$$2 \cdot 3^k \equiv \rho \pmod{7^E}$$

for $1 \leq E \leq 16$.

The logarithmic coordinate agreed at all tested precisions, the local slope remained a unit, and the principal residue class was unique modulo 7^{15} at the final precision. Specifically, if m_* denotes the final principal coordinate, then

$$m_* + 7^{15}j$$

produces the same orbit residue modulo 7^{16} for every integer j , while perturbations by 7^{14} change the residue. This is the correct uniqueness statement: uniqueness of the principal coordinate modulo 7^{E-1} .

11 What is proved and what remains computational

It is important to distinguish the algebraic statements from the finite-data audits.

1. The Schur identity, the rank-one correction, the quotient dimensions, and the adjugate kernel memberships are exact finite-dimensional linear algebra statements for the audited matrix.
2. The factor-17 scaling law is exact for the audited source data and its controlled counterfactuals.
3. The principal-unit linearization and the formula

$$B(\rho) = \rho(729 - 1)/7 \pmod{7}$$

are algebraic statements, and the computation confirms them for all $\rho \in \mathbf{F}_7^\times$.

4. The logarithmic formula is a standard 7-adic consequence in the unit neighborhood where the logarithm converges, and its agreement with the reconstructed coordinate was verified through 7^{16} .
5. No claim is made here that the particular integer matrix construction, the terminal source values, or the observed 17-adic localization extends automatically to other unrelated matrices or source tables.

12 Conclusion

The exact audits reveal a coherent two-prime structure.

At $p = 2$, the Schur extension contributes a rank-one correction which removes exactly one dimension from the retained row and column spaces. The distinguished adjugate directions lie in the corresponding core kernels. The boundary determinant 2 is responsible for the non-unimodular behavior of the Schur transformation.

At $p = 17$, the common source factor is localized to the terminal data and transfers linearly through the Schur source layer. Common rescaling changes amplitude but not the projective source ratio.

At $p = 7$, that projective ratio determines a canonical exponent coordinate. Writing $k = k_1 + 6m$ converts the multiplicative orbit into the principal-unit coordinate 729^m . The local Hensel slope is universally

$$B(\rho) = \rho \frac{3^6 - 1}{7} \pmod{7},$$

which is always a unit for $\rho \in \mathbf{F}_7^\times$. Hence the lift digit is unique at every level. For the original source ratio $\rho \equiv 6 \pmod{7}$, the slope is exactly 1, explaining the experimentally persistent recurrence.

The principal coordinate is simultaneously characterized recursively and logarithmically:

$$m = \frac{\log(\rho/6)}{\log(729)}$$

for the original chart, with the corresponding general chart obtained by replacing 6 with $2 \cdot 3^{k_1}$. The finite computations through 7^{16} are consistent with this exact 7-adic description.

Theorem 12.1 (Principal-coordinate theorem for the audited orbit). *Let $\rho \in \mathbf{Z}_7^\times$ satisfy $\rho \not\equiv 0 \pmod{7}$, and let $k_1 \pmod{6}$ be determined by $2 \cdot 3^{k_1} \equiv \rho \pmod{7}$. Define $m \in \mathbf{Z}_7$ in the principal-unit chart by*

$$\frac{\rho}{2 \cdot 3^{k_1}} = 729^m.$$

Then the exponent solutions to

$$2 \cdot 3^k = \rho$$

in the 7-adic unit group are exactly

$$k = k_1 + 6m.$$

The digit-by-digit Hensel lift of k is unique at every level, and its principal digits are the canonical base-7 digits of m . In the original chart $k_1 = 1$ this becomes

$$m = \frac{\log(\rho/6)}{\log(729)}.$$

Proof. The decomposition of $\mathbf{Z}_7^\times$ into the residue class modulo 7 and the principal-unit subgroup gives the unique $k_1 \pmod{6}$. After factoring out $2 \cdot 3^{k_1}$, the equation becomes $729^m = \rho/(2 \cdot 3^{k_1})$. Since $729 = 1 + 7 \cdot 104$ is a principal unit with nonzero first logarithmic coefficient, the map $m \mapsto 729^m$ is an analytic isomorphism from $\mathbf{Z}_7$ onto the closed subgroup generated by 729. The logarithm linearizes this map, giving

$$m = \frac{\log(\rho/(2 \cdot 3^{k_1}))}{\log(729)}.$$

The nonzero derivative modulo 7 gives unique Hensel lifting at every finite level, and the recurrence $m_{e+1} = m_e + t_e 7^{e-1}$ is precisely the canonical base-7 expansion of m . □ □

Computational provenance

The paper consolidates the exact outputs of Experiments 164U through 217, with the theorem-oriented checkpoints being Experiments 165R–180 (Schur geometry and terminal source arithmetic), 181–184 (17-adic localization and counterfactual scaling), 185–200 (3- and 7-adic terminal behavior), 201–210 (long Hensel reconstruction and principal-coordinate digits), 211–215 (local linearization, logarithmic coordinate, inverse/uniqueness checks), and 216RRR–217 (projective/content decoupling and the universal mod-7 slope law).